

COMPETITOR GUIDE - v1.0.0

RANK :: CHALLENGE

Active Directory & Windows Admin Challenges

10 CHALLENGES - 2,650 POINTS - POWERSHELL ONLY

A progressive, command-writing Windows administration challenge that takes you from DNS and firewall rules all the way to passwords and Group Policy. Each level unlocks the next. Write the command, earn the points, climb the admin ranking.

Generated 2026 - RANK::CHALLENGE v1.0

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What is Rank Challenge?

Rank Challenge is a command-writing competition for Windows administrators. Every level hands you a small admin task. Your job is to write the PowerShell that performs it - and the platform grades your command automatically.

There are 10 challenges worth a combined 2,650 points. They form a single progression: each level you solve unlocks the next, and your total score places you on the admin ranking.

Every challenge is PowerShell

The whole track is PowerShell, using the standard Windows admin modules. You write commands in .ps1 files inside an in-browser editor. Topics span core directory and server administration:

- DNS - host (A) and alias (CNAME) records with the DnsServer module
- Firewall - creating, changing, and enabling rules with NetSecurity
- Passwords - resetting accounts and hardening the domain policy
- Group Policy - creating, linking, and configuring GPOs
- Accounts - identities, groups, and remediation with ActiveDirectory

You only need to know PowerShell

No programming is required. If you can read and write a PowerShell cmdlet with its parameters, you have everything you need.

How grading works

When you press Submit, the server checks your command for the cmdlet and parameters that level requires (for example: "uses New-NetFirewallRule", "sets -Action Allow", "uses Set-GPRegistryValue"). All checks must pass to clear the level.

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Install Node.js

Rank Challenge runs on Node.js (which includes npm). You need Node 18 or newer; the latest LTS release is recommended.

Step 1 - Check if you already have it

Open a terminal (Terminal on macOS/Linux, or PowerShell / Command Prompt on Windows) and run:

```
node -v  
npm -v
```

If both print a version (and node -v is v18 or higher), you're ready. Otherwise install or update Node below.

Step 2 - Download & install

Official installer (pick LTS): <https://nodejs.org/en/download>

Run the installer, accept the defaults, then reopen your terminal and rerun node -v to confirm.

```
# macOS (Homebrew)  
brew install node  
# Windows (winget)  
winget install OpenJS.NodeJS.LTS  
# Debian / Ubuntu  
sudo apt-get install -y nodejs npm
```

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Run the Challenge

With Node installed, you start the platform from the project folder in three commands.

Step 1 - Open the project

```
cd path/to/ad
```

Step 2 - Install dependencies (first time only)

```
npm install
```

This installs the frontend and backend packages (npm workspaces). It only needs to be done once.

Step 3 - Start it

```
npm run dev
```

This launches the frontend (<http://localhost:5173>) and the backend (<http://localhost:3001>) together.

Step 4 - Open & log in

Open <http://localhost:5173>, click Continue, then enter your 6-digit access code. A successful login starts a 1-hour session.

Login codes are issued by the 2FA service your organiser runs (<http://localhost:3000>). Each code is single-use and expires quickly.

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The 10 Challenges

All ten levels are PowerShell. Solve them in order - each unlocks the next.

#	TITLE	WHAT YOU'LL FIX	POINTS
01	Create a DNS Record	Add-DnsServerResourceRecordA	100
02	Alias a Host	Add-DnsServerResourceRecordCName	150
03	Open a Firewall Port	New-NetFirewallRule (inbound 443)	200
04	Tighten a Rule	Set-NetFirewallRule -Action Block	250
05	Enable a Rule	Enable-NetFirewallRule	250
06	Reset a Password	Set-ADAccountPassword	300
07	Harden Password Policy	Set-ADDefaultDomainPasswordPolicy	300
08	Create a GPO	New-GPO	350
09	Link a GPO	New-GPLink to an OU	350
10	Policy Registry Setting	Set-GPRegistryValue	400

Each level tells you exactly what to do

Every challenge has a description, a list of objectives, and optional hints. The objectives map directly to the checks the grader runs.

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Scoring & Rules

Points

Each level is worth the points shown above, from 100 (Level 01) up to 400 (the finale) - 2,650 in total. Points are banked the moment all of a level's checks pass.

Hints cost points

Revealing a hint subtracts a fixed penalty from that level's score (around 15 points). Use them only when stuck.

Unlock chain

Levels unlock in sequence: clear Level 01 to open Level 02, and so on to the finale. Your progress is saved within your session.

Session window

A login is valid for 1 hour. When the timer in the top bar runs out you return to the login screen and need a new code.

Submission limit

Submissions are rate-limited to 20 per minute to keep things fair - no penalty, you just wait a moment if you hit it.

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The Interface & Finishing

Top bar

- Server status - confirms the backend is reachable
- Session timer - time left in your 1-hour window
- Score & progress - your points and solved/total count
- Challenge Map - jump back to the level grid
- Finish - end your attempt

Editor & results

The left panel is the script editor (your .ps1 file). The right panel is a results console - press Submit to grade your command and see which checks passed.

Use Submit to grade your answer, Hint to reveal a hint (costs points), and Reset to restore the original starter script (.ps1 file).

Finishing the challenge

When you're done, press Finish in the top bar to see a summary of how many levels you solved and your final score, then confirm to end the session.

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Tips & Quick Reference

Strategy

- Read the objectives first - they mirror exactly what the grader checks.
- Cmdlets and parameters are case-insensitive, but spell them exactly - Get-ADUser, -Filter, -Identity.
- Save hints for later - try the fix yourself first; hints cost points.
- Submit early, submit often - failed checks tell you what's still wrong.
- Reset if you get lost - it brings back the original starter.

Quick command reference

```
node -v          # check Node version (need 18+)
npm install      # one-time dependency install
npm run dev      # start frontend + backend
Ctrl + C        # stop both servers
```

Good luck - fix the commands, earn the points, climb the ranking.